Digital Pendulum System

INSTALLATION AND COMMISSIONING

33-005-0M5

MATLAB 5 Version

Feedback

Feedback Instruments Ltd, Park Road, Crowborough, E. Sussex, TN6 2QR, UK.
Telephone: +44 (0) 1892 653322, Fax: +44 (0) 1892 663719.
email: feedback@fdbk.co.uk website: http://www.fbk.com

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Notes

THE HEALTH AND SAFETY AT WORK ACT 1974

We are required under the Health and Safety at Work Act 1974, to make available to users of this equipment certain information regarding its safe use.

The equipment, when used in normal or prescribed applications within the parameters set for its mechanical and electrical performance, should not cause any danger or hazard to health or safety if normal engineering practices are observed and they are used in accordance with the instructions supplied.

If, in specific cases, circumstances exist in which a potential hazard may be brought about by careless or improper use, these will be pointed out and the necessary precautions emphasised.

While we provide the fullest possible user information relating to the proper use of this equipment, if there is any doubt whatsoever about any aspect, the user should contact the Product Safety Officer at Feedback Instruments Limited, Crowborough.

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Refer to accompanying documents

CAUTION - RISK OF ELECTRIC SHOCK

CAUTION - ELECTROSTATIC SENSITIVE DEVICE

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All major changes are incorporated into up-dated editions of our manuals and this manual was believed to be correct at the time of printing. However, some product changes which do not affect the instructional capability of the equipment, may not be included until it is necessary to incorporate other significant changes.

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Where components are of a ‘Safety Critical’ nature, i.e. all components involved with the supply or carrying of voltages at supply potential or higher, these must be replaced with components of equal international safety approval in order to maintain full equipment safety.

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1. Equipment type
2. Component value
3. Component reference
4. Equipment serial number

Components can often be replaced by alternatives available locally, however we cannot therefore guarantee continued performance either to published specification or compliance with international standards.

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This equipment, when operated in accordance with the supplied documentation, does not cause electromagnetic disturbance outside its immediate electromagnetic environment.

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1 INTRODUCTION

This manual contains a description of the components of the Digital Pendulum System, assembly instructions and test procedures. It is intended to provide the means for validating the correct operation of the system, ***before*** proceeding to more advanced functions and demonstrations using control software with MATLAB / SIMULINK.
The procedures described in this section of the Pendulum manual will ensure correct:

- Assembly of the system
- Switch and Jumper settings of the PCL-812 interface card
- Cable connections between the PC and the Pendulum

1.1 REQUIRED EQUIPMENT

The following software and hardware equipment together with documentation are required to use the system.

- **IBM Compatible 486DX or Pentium PC** with minimum configuration:
  16 Mb of memory, Math co-processor (included in 486 or Pentium processor)
  CD-ROM
  Hard Disk with 100 Mb available after MATLAB installed
  SVGA screen and mouse.
  DOS version 5 or later, Windows 95 or Windows NT
- **MATLAB** Version 5.3 with SIMULINK, Control Systems and Signal Processing toolboxes plus Fuzzy Logic NCD.
- **Advantech** PCL-812PG Enhanced Multi-Lab Card or the equivalent such as RT-DAC to install in the PC with 1 x 20 way and 1 x 40 way ribbon cables
- **Feedback-** Pendulum System consisting of Digital Pendulum Mechanical Unit(33-200) and Digital Pendulum Controller Unit(33-201).
- **CD-ROM** (33-930) with Feedback's Pendulum Control Software,
- **Set of 6 manuals:**
  33-000-0 - Feedback MATLAB Based Control Products Software Installation Guide
  33-005-0M5 - Pendulum System - Installation and Commissioning
  33-005-1M5 - Pendulum System - Getting Started
  33-005-2M5 - Pendulum System - Reference Manual
  33-005-3M5 - Pendulum System - External Interface
  33-005-4M5 - Pendulum System - Teaching Manual 1
  33-005-5M5 - Pendulum System - Teaching Manual 2

1.2 FEEDBACK LTD SUPPLIED ITEMS

The following items are supplied by Feedback under the following product codes

33-200	Digital Pendulum Mechanical Unit
33-201	Digital Pendulum Controller
33-930	Digital Pendulum Control System Software pack including Advantech card
33-930I	Digital Pendulum Control System Software pack excluding Advantech card
33-005	Contains 33-200 , 33-201 , 33-930 (Composite product)
33-005I	Contains 33-200 , 33-201 , 33-930I (Composite product)

2 MECHANICAL ASSEMBLY

Carefully unpack the items, and remove all packaging and securing material. You should have the following items:

1) Track for the cart to move on, with motor attached, and toothed belt
2) Pendulum cart with mounting points for the two pendulum arms and attached ribbon cable
3) Two pendulum arms with weights attached
4) Two legs each with two adjustable feet, with mounting plates for securing to the track
5) 1 x 2mm and 1 x 6mm Allen keys

Refer to Fig 2.1 below

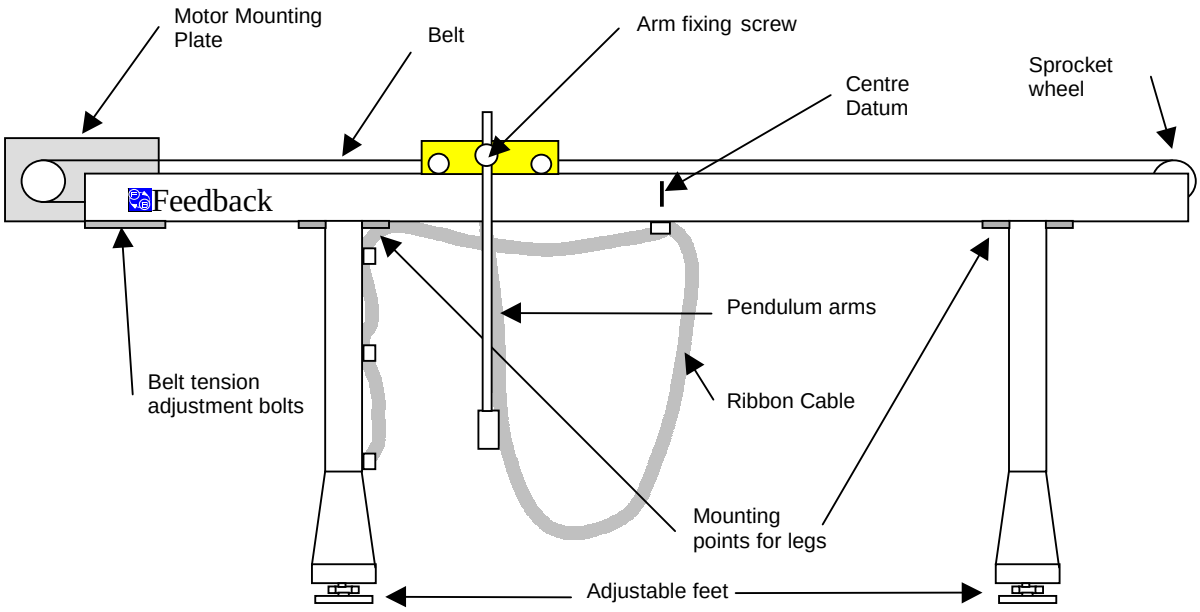


Fig 2.1: The pendulum system assembled

Place the Digital Pendulum Mechanical Unit on a level surface with the motor on the left viewed from the front and allow sufficient room for the controller box , PC and mouse. Ensure that the pendulum is clear of any obstructions when executing a complete revolution, and is not able to make contact with any observers.

Using the 6mm Allen key provided, remove the 8 bolts attached to the track at the leg mounting positions, and secure the legs to the track with these bolts.

Use the adjustable feet (it is only necessary to adjust one of the four) to ensure that the pendulum system is stable on the surface. Slacken off the adjustable foot locking nut and rotate the foot until the system is perfectly stable. Tighten the locking nut.

Insert one of the pendulum arms into one of the two mounting positions on the cart from the bottom and loosely tighten the securing screw on to the flat part of the pendulum arm with the 2mm Allen key provided. Push the arm upwards until the end of the flat is reached and tighten the securing screw. Repeat for the other pendulum arm on the other side of the cart

Ensure the pendulum cart is free to move on the track and that the drive belt is attached to the cart at both ends and goes over the sprocket wheels at each end of the track. If the belt is slack, use the 6mm Allen key to slacken off the belt tension adjustment bolts and move the motor mounting plate until the belt is taut. Tighten the adjustment bolts.

Finally, clip the ribbon cable to the pendulum left hand leg using the attached clips.

3 SWITCH AND JUMPER SETTINGS FOR THE PCL- 812 CARD

The PCL-812 contains components which can be damaged by static electricity. Before removing the PCL-812 board from its conductive packaging, touch your hand on a grounded metal object such as the case of your PC. Avoid touching the components on the board or the plated edge connectors.

Before installing the PCL-812PG interface card into your computer make sure that the switches and jumpers shown in Figure 3.1 are set correctly, as detailed in this section.

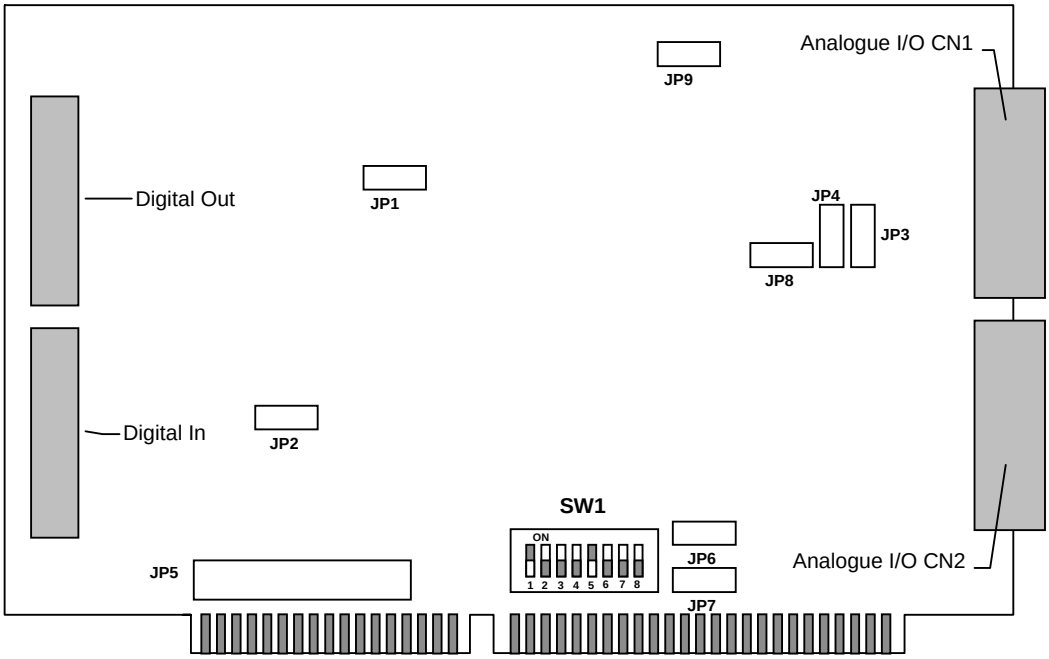


Figure 3.1: 16 bit PCL-812PG Interface Card - Switch and Jumper positions

Note: The board shown above is a 16 bit short board. Some users may have the 8 bit full length PCL- 812 board for which the SW1 switch settings are different. If you have this board refer to the settings in the following section for the 8 bit board.
Other settings and cable connections are the same for both boards.

3.1 SWITCH SETTINGS - 16 BIT BOARD

Switch SW1 controls the base address for I/O transfer (switches 1-6), and the CPU wait states (Switches 7-8). They should be set as follows.

(Note that the Pendulum software always refers to the decimal settings, whereas the manual on the PCL-812 board refers to hexadecimal settings. For convenience, both decimal and hex addresses are shown in the table below).

SW1 Base Address Selection

I/O Address Range (hex and decimal)	**SW1 Switch position**					
	1	**2**	**3**	**4**	**5**	**6**
200-20F (512-527 Dec)	OFF	ON	ON	ON	ON	ON
210-21F (528-543 Dec)	OFF	ON	ON	ON	ON	OFF
220-22F (544-559 Dec) *	OFF	ON	ON	ON	OFF	ON
230-23F (560-575 Dec)	OFF	ON	ON	ON	OFF	OFF
300-30F (768-783 Dec)	OFF	OFF	ON	ON	ON	ON
3F0-3FF (1008-1023 Dec)	OFF	OFF	OFF	OFF	OFF	OFF

***** Factory default setting

Note that some video cards or floppy disk drives use the higher base addresses such as 1008 (Hex 3F0), and should be avoided if possible.

SW1 Wait State Selection

	SW1 Switch Settings	
Time Delay	7	8
0 *	ON	ON
2	OFF	ON
4	ON	OFF
6	OFF	OFF

***** Factory default setting

3.2 JUMPER SETTINGS

The Jumpers on the PCL-812 board should be set in the positions indicated below.

JP1 - Trigger Source Selection

	JP1
INT	O
TRG	O
EXT	O

JP2 - Counter Input Clock Selection

	JP2
INT	O
CLK	O
EXT	O

JP3, JP4 - D/A Reference Selection

	INT	VREF	EXT	
JP3	O	O	O	D/A1
JP4	O	O	O	D/A2

JP5 - IRQ Level Selection

2	3	4	5	6	7	X
O	O	O	O	O	O	O
O	O	O	O	O	O	O

JP5

JP6, JP7 - DMA Channel Selection

1	3	X
O	O	O
O	O	O

JP6

1	3	X
O	O	O
O	O	O

JP7

JP8, JP9 - D/A Internal Reference, A/D Maximum Input Voltage Selection

	10	5	
JP8	O	O	O
JP9	O	O	O

Depending on the specification and configuration of your PC you may need to change the Base Address and Wait State settings.

None of the other settings need to be changed.

3.3 SWITCH SW1 SETTINGS FOR 8 BIT FULL LENGTH PCL - 812 BOARD

I/O Address Range (hex and decimal)	SW1 Switch position					
	1	**2**	**3**	**4**	**5**	**6**
200-20F (512-527 Dec)	ON	ON	ON	ON	ON	X
210-21F (528-543 Dec)	ON	ON	ON	ON	OFF	X
220-22F (544-559 Dec) *	ON	ON	ON	OFF	ON	X
230-23F (560-575 Dec)	ON	ON	ON	OFF	OFF	X
300-30F (768-783 Dec)	OFF	ON	ON	ON	ON	X
3F0-3FF (1008-1023 Dec)	OFF	OFF	OFF	OFF	OFF	X

Note: An “X” in the table above means that this switch is not used.

4 INSTALLATION OF THE PCL-812 CARD AND CABLES

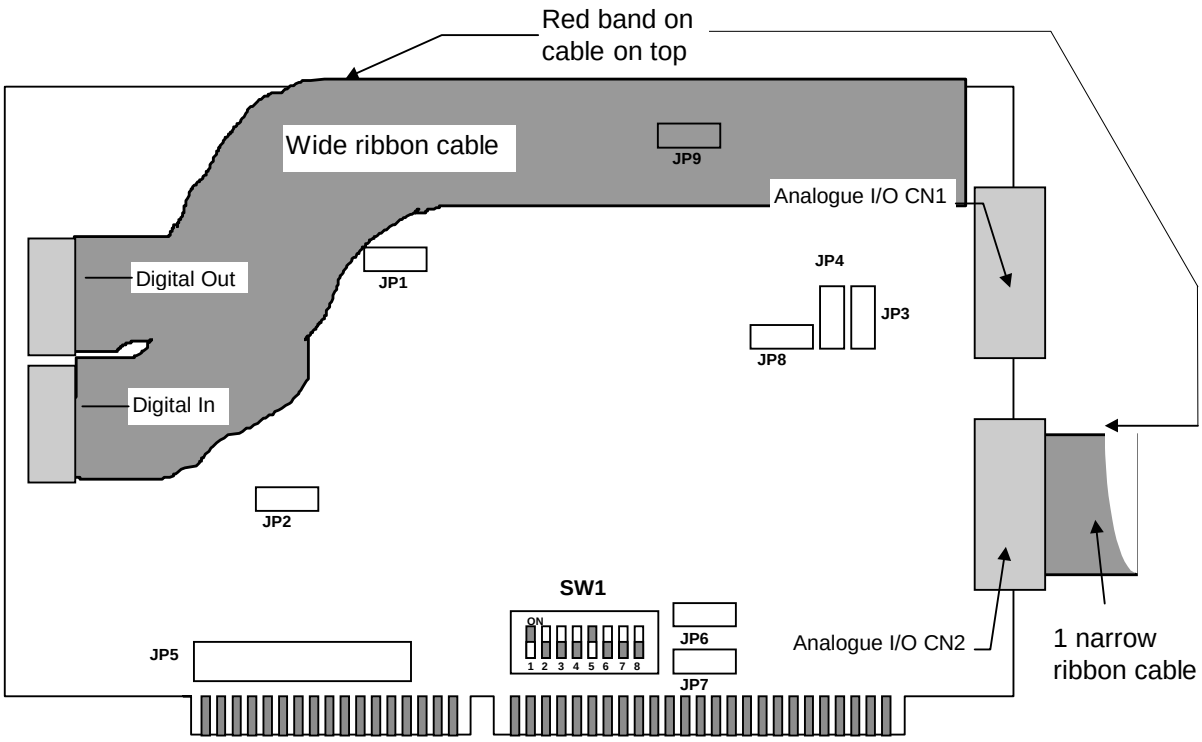


Figure 4.1: PCL- 812 Interface Card Ribbon Cable connections

Two slots are required at the rear of the PC - one for the two 20 way analogue port sockets, and one for the 40 way socket at the end of the wide ribbon cable for the digital ports. Use screws to secure the card and ribbon socket plates to the casing of the PC.

Carefully install the interface card in a spare slot on the PC motherboard (or in a slot on the passive backplane if your PC has this type of architecture - common with some 19 inch rack mounted PC's). If in any doubt as to the correct procedure consult the documentation supplied with the PC.

Turn OFF and disconnect the mains power supply
BEFORE opening the computer

Before opening the computer it is important to ensure that the computer is switched off and that the mains power supply is not connected.

Once you have opened the computer, check to see if there are any other cards installed, for example a network or sound card. It is important to determine the addresses of any such cards as your Pendulum System will not function correctly if there is another card using all, or part of, the address space assigned to the PCL-812PG interface card.

Following the installation of the interface card into your PC, you are now ready to close the PC case and connect the ribbon cables to the 33-201 Digital Pendulum Controller.

Refer to Figures 4.2 and 4.3 below for details.

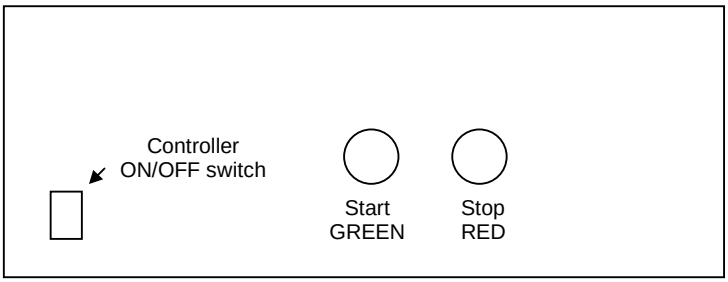


Fig. 4.2: Front view of 33-201 controller

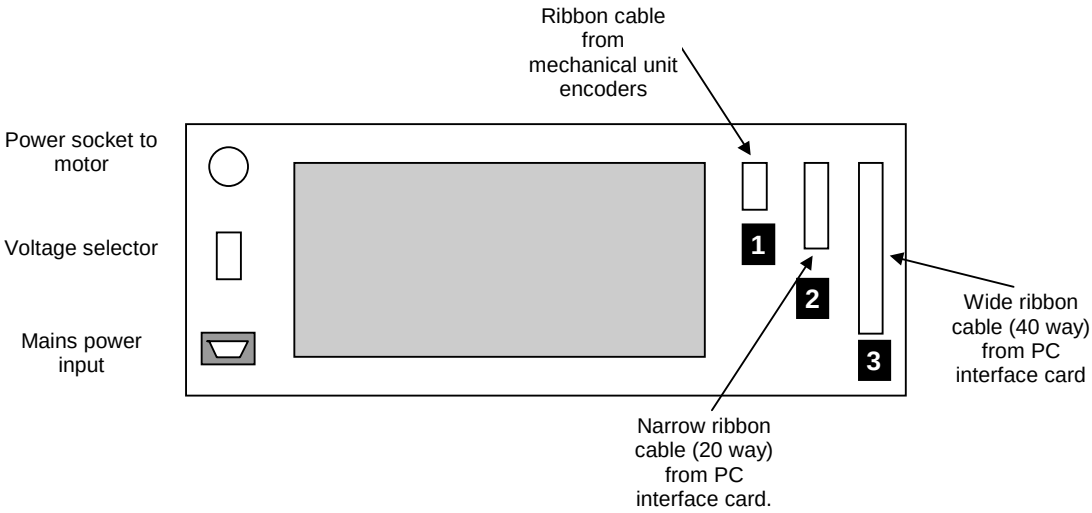


Fig. 4.3: Rear view of 33-201 controller

Connectors and Switches

The sockets for the ribbon cables are on the back of 33-201 unit. The 20-way connector (No. 2) connects to CN2 of the PC interface card. This is the lower connector on the interface card (i.e. closest to the motherboard of the PC). The 40 way (No. 3) connector on the 33-201 connects to the 40 way connector on the PC interface card. The smallest connector on the 33-201 (No. 1) connects to the ribbon cable from the mechanical unit encoders.

Voltage selector

Set the voltage selector for the correct voltage in your area.

Mains power

Connect the mains power cable to the 33-201 and plug in to your mains supply.

Switch on

Switch on the mains supply to the 33-201 and your PC. Switch the 33-201 illuminated power switch on. If this switch does not light up check the mains power cable and mains power supply.
Do not at this stage press the round green start button.

It is important that the area of operation is free of obstruction and that users and observers are warned to keep clear of the immediate vicinity of the system for safety reasons.

5 INSTALLATION AND TESTING OF THE RT-DAC CARD

See: the *RT-DAC multi I/O board User's manual* for installation, and the *RT-DAC testing software* for testing the card.

Notes

6 RTK AND MATLAB SOFTWARE INSTALLATION

See the *Installation Guide 33-000M5 : Feedback software installation programs for digital control equipment working with MATLAB 5.3 under Windows environment* for details.

Notes

7 COMMISSIONING AND TEST SOFTWARE

7.1 STARTING THE SOFTWARE

Commissioning/Test software is supplied to check the operation of the Pendulum System, before using the MATLAB driven control software.

If you use the RT-DAC card please remember to use the RT-DAC Testing Software to check the data flow between a PC and the Pendulum System.

Windows 95 installation

Find the Pendulum Commissioning /Test software file on CD-ROM supplied:

From the Windows 95 **Start** menu click on **RUN** and use the browse feature to locate the **PENDTST.EXE** program on CD-ROM drive:

Windows 3.1 installation(Ignore this if you are using Matlab version 5.1 or above)

Insert the Pendulum Commissioning / Test software disk in drive A:

Start **File Manager** and click on the **PENDTST.EXE** program on drive A:. Click on **RUN** from the File manager **File** menu.

DOS installation

Insert the Pendulum Commissioning / Test software disk in drive A: Type in the following DOS commands.

>**A:**

>**PENDTST**

7.2 SOFTWARE OPERATION

After starting the Pendulum commissioning/test software you will see the following screen presented.

Fig 5.1. Start screen

- Click **OK.** The following screen appears, ***only if*** *the card is not found at the factory default Base address of 544.*

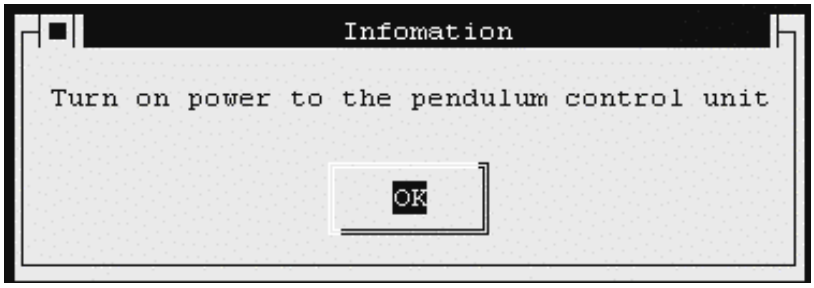
Fig. 5.2: PCL-812 Base address Selection

- Use the mouse to select the base address of the PCL-812 interface card which has been set and checked in accordance with the procedure detailed in Chapter 3 of this manual.

  If you do not know the Base Address, clicking on **Locate** will identify and set it automatically. (Note that **Locate** can only be used if there is only ***one*** PCL-812 board installed in your PC).

- Having selected the correct base Address, click on **Close**.

The following dialogue box appears.



- Remove the drive belt from the motor sprocket wheel.

- View the pendulum so that the motor is on the left side of the rail and the Feedback logo on the front.

- Turn on the power to the pendulum control unit and click OK.

The test of the motor must be done with the drive belt removed from the motor sprocket wheel to avoid the cart hitting the end stops.

The following screen appears.

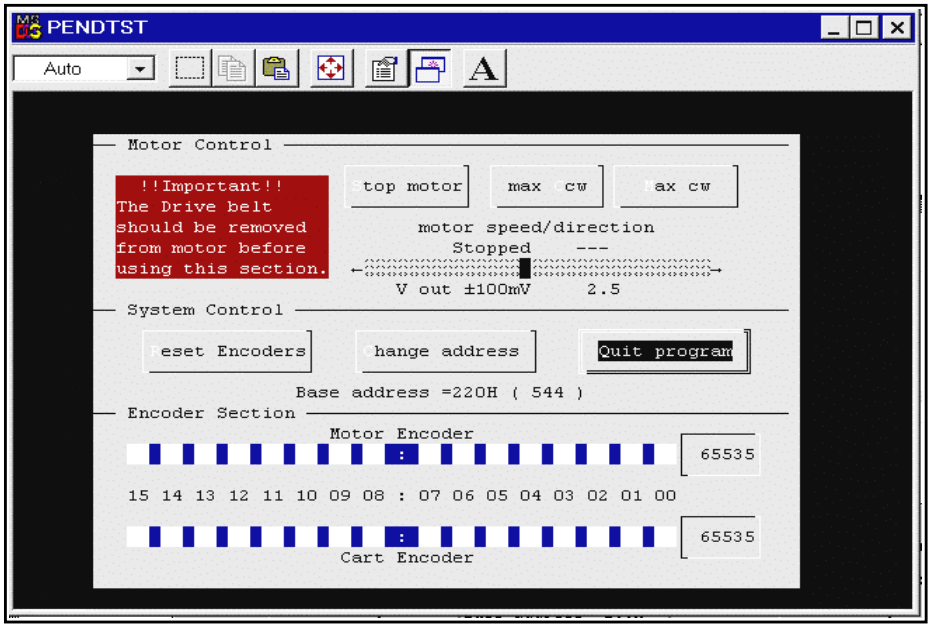


Fig. 5.3: Main screen of PENDULUM Commissioning and Test Software .

7.3 DESCRIPTION OF SCREEN CONTROLS AND DISPLAYS.

This screen allows you to interact with the PENDULUM System, change the speed or reverse direction of the motor by changing the applied voltage, and display the outputs from the two rotary encoders, which measure the angular displacement of the pendulum and the linear displacement of the cart.

Max cw	Clicking this button applies the maximum voltage to the motor to rotate it in a clockwise direction, from the viewpoint specified.
max Ccw	Clicking this button applies the maximum voltage to the motor to rotate it in a counter clockwise direction, from the viewpoint specified.
Stop motor	Clicking this button will stop the motor.
Change address	Changes the base address of the PCL-812 board. The base address currently used is shown under this button in decimal format.
Reset encoders	Click on the reset encoders button sets the rotary encoders to zero. Subsequent encoder outputs are then angular deviations relative to the new set positions. The outputs for the motor and cart encoders, which measure the linear cart displacement and the pendulum angular displacement, are shown in binary format at the bottom of the screen
Quit Program	Exits the program.
Motor speed/direction	Dragging the slider bar to the left or right allows the motor speed to be varied in a clockwise or counter clockwise direction, in a number of discrete steps. When the slider bar is in the centre of the slider the motor is stopped, with an output voltage of 2.5 ± 100mv.

HOT KEYS
Note that as an alternative to using the mouse, the functions displayed on the screen can be activated by typing ALT + the highlighted letter displayed in reverse video on the screen. Thus, for example, to reset the encoders, press ALT + R

8 TEST SEQUENCE

When you have complete the system installation, execute the following test sequence to ensure that the system is functioning properly.

- Start PENDTST commissioning software, and check that the base address is correctly set.
- Switch on the mains power and the switch on the front of the 33-201 pendulum control unit.
- Click on **Stop motor.** Start the sequence in the table below.

ACTION	RESULT
1. Disconnect the drive belt from the motor sprocket. Place the cart in the centre of the rail, lining up the positioning marks on the cart and the rail. Click **Reset Encoders** Slowly rotate the pendulum in a clockwise direction (as viewed from the specified viewpoint).	Observe the change in the **cart encoder** display in the bottom part of the screen. The encoder output is shown as a number.
2. Move the cart as far to the left as possible. Click **Reset Encoders**. Move the cart slowly as far to the right as possible. Move the cart slowly as far to the left as possible.	Observe the change in the **motor encoder** display in the bottom part of the screen. The encoder output is shown as a number. The encoder output should change from around zero to a value between 13000 and 14000 The motor encoder output should return to a value close to zero.

ACTION	RESULT
3. Click the **Stop motor** button Drag the Motor control slider at the top of the screen a small amount to the right. Repeat the above, dragging the Motor control slider the maximum amount to the right.	Establish that the motor is turning clockwise from the specified viewpoint. The motor is turning clockwise at its maximum rate.
4. Click the **Stop motor** button Drag the Motor control slider at the top of the screen a small amount to the left. Repeat the above, dragging the Motor control slider the maximum amount to the left.	Establish that the motor is turning counter clockwise from the specified viewpoint. The motor is turning counter clockwise at its maximum rate.

Always be prepared to press the round red STOP button to cut the power to the pendulum if the system exhibits erratic or unexpected behaviour.